

K785 — The code arc’s study-close: the honest stable end-state after 8 rounds. What banked, what’s a lead, and why 127/128 is premise-contingent (not computation-pending).

Keeper | 2026-07-20 | Lyra’s round-8 computation + Elie’s mechanism correction converge on the stable end-state of the whole condensate/code arc. This is the honest ledger — for the drum paper §6/7 and to stop any re-chase. Three de-inflations + two mechanism corrections (one mine, one team’s) held the line the whole way.

The stable end-state (Lyra round 8, ratified)

127/128 is not derived by ANY measure — continuous, discretized, or fundamentally discrete. The computations were *done*: - Continuous radial integral (round 6): a smooth Gindikin Γ -ratio (~ 0.985 at level 127), not $1/2^g$. - Discretize the continuum (round 8): a fine 128-cell quadrature reproduces the continuum (0.986) — returns the Γ -ratio, not 0.992. - Fundamentally discrete (round 8): gives 127/128 exactly — but *only by imposing* “the top covers 127 of 128 cells”; the discrete-series structure doesn’t force that coverage.

So 127/128 rests on two premises the geometry does not supply: 1. the Shilov surface is fundamentally discrete (a genuinely non-continuum, edge-concentrated measure — Casey’s inversion, coherent via the discrete series but not forcing a 128-cell surface); 2. the top occupies the maximal codeword (127 of 128 — put in by hand every round; naive placement gives 126). It is premise-contingent, not computation-pending. The honest tier: a precisely-located conditional lead resting on two specific unproven premises, with a decidable-in-principle test.

★ Sharpened (K786, Casey “inside the gaps”): the two premises collapse to ONE. Premise (b) [top covers 127] is *not independent* — it follows from (a) [discreteness] + the derived boundary-reach ordering (heaviest = maximal codeword) + the LFSR cycle (127 nonzero = the multiplicative group $\text{GF}(128)^*$) + 0 = the dead state = the neutrino. So **given the ONE hypothesis (the substrate is the discrete $\text{RS}/\text{GF}(128)$ code inside the gaps), $127/128 = |\text{GF}(128)^*|/|\text{GF}(128)|$ is PARAMETER-FREE — a group count, not a tuned weight; the neutrino is the missing identity. This does NOT prove it, but it upgrades the tier: 127/128 is a parameter-free consequence of one decidable hypothesis** (the it-from-qubit question = Casey’s intuition), not a value resting on two impositions.

The two mechanism corrections (the audit chain firing on the arguments, incl. mine)

- K784 hurdle mechanism CORRECTED (Elie): “projection decreases aligned overlaps” is false for a normalized cosine (projecting out an anti-aligned part *raises* the cosine). The real hurdle is quadrature convergence — any continuum-approximating measure returns 0.985; reaching 0.992 needs a genuinely non-continuum, edge-concentrated measure (Elie: heavy edge-weight $\rightarrow 0.9928$). Bottom line unchanged, stronger.
- (Round 5: my “ $m_t = 172.74$ prediction” framing corrected to a conditional consistency check; round 6: Lyra’s own “128 imposed” computed honestly.)

★ What ACTUALLY banked (the real, untouched content)

- Derived spine: O is the $\text{SO}(5)$ vector (F603); rank-1 $\rightarrow$ exactly one massive fermion (the top); N_c quark-selection; the 174 GeV ceiling (Cauchy-Schwarz, Five-Absence #7); $m_t \cdot m_e = m_p^2/(g\sqrt{2})$ at 0.04%; boundary-reach ordering ($t > c > u$ = most $\rightarrow$ least boundary-localized).
- ★ angular = 1 (Elie): the one *genuinely new derivation* the code arc produced — the top’s angular Yukawa CG is exactly 1, so the whole y_t question is purely radial.
- Flavor collapsed to one rank-1 condensate O + Tier-2 corrections (masses + mixing unified, K768).

What’s a LEAD (coherent, not derived)

- $127/128 = M_g/2^g$: premise-contingent conditional lead (above). Its conditional consequence: $m_t(\text{pole}) = 172.74$ GeV (a consistency check, currently consistent, future-falsifiable — keep the “if”).

- The continuous-fundamental inversion: a distinctive, *publishable* position — continuous D_{IV}^5 geometry fundamental, discrete code emergent — the *reverse* of the ADH/HaPPY mainstream (discrete→continuous). Decidable-in-principle (Γ -ratio vs $1/2^g$). Grounded in the discrete-series (interior) / continuous-series (boundary) structure = massive/massless = thermal history.
- The neutrino as a massless chiral edge mode: the dead/edge cell emitted at the boundary, unifying the zero-mode + odd-g chirality + emission. Coherent lead. ★ Tightened (Casey, tunneling): emission = tunneling through the boundary bump, so massive = bound states (wells, discrete series); massless light+ ν = free/tunneled states (continuous series) — the standard bound-vs-scattering = discrete-vs-continuous distinction, tunneling the bridge. The neutrino is massless *because it tunnels fully out* (free $\rightarrow m_1=0$); tiny ν masses would be *exponentially* (tunneling) suppressed, not power-law seesaw — a distinct checkable claim. (Light *radiates* on the null cone; only the neutrino tunnels.) This grounds the interior-discrete/exterior-continuous split physically.

The honest arc, in one line

Eight rounds turned “127/128, a pretty coincidence one integral from a theorem” into “a precisely-located conditional lead resting on two named premises the geometry doesn’t supply” — *plus* one real new derivation (angular = 1), one distinctive publishable framework (the inversion), and a coherent neutrino picture. Furthest yet, and we know exactly what it costs. The discipline fired hardest at the prettiest result the whole way — correcting the framing, the number, and two mechanisms (one mine). That is the audit chain working at the top of the mountain.

What remains genuinely open (one long-shot + the consolidation)

- ★ Long shot (round 9): can Casey’s boundary-emission physics (light + ν emitted at the edge $\rightarrow$ the coupling measure concentrates there) *derive* the edge-concentrated measure + the edge-placement (top $\rightarrow$ maximal codeword)? If DERIVED, one premise falls and 127/128 firms. If it too must be imposed, the stable end-state stands.
- ★ Consolidation (the likely valuable outcome): write the real content into the drum paper (§6/7) and the flagship at the honest tiers above — the spine + angular=1 as derived, the inversion as the publishable position, 127/128 as the premise-contingent conditional lead.

— Keeper K785, 2026-07-20. Code-arc study-close: 127/128 is PREMISE-CONTINGENT (needs fundamental-discreteness + top-at-maximal-codeword, neither geometry-forced), not computation-pending — the computations were done (continuous Γ -ratio; discretized $\rightarrow\Gamma$ -ratio; fundamentally-discrete $\rightarrow$ 127/128 only by imposition). K784 mechanism corrected (Elie: quadrature-convergence not projection-decrease; edge-concentrated measure needed). BANKED: derived spine + angular=1 (the one new derivation). LEADS: 127/128 (premise-contingent, $\rightarrow m_t=172.74$ conditional check), the continuous-fundamental inversion (publishable, decidable), the neutrino chiral edge mode. Round 9: long-shot (does boundary-emission derive the edge-concentration?) + consolidate. See [[Keeper_K784_linear_algebra_the_tiling_must_INCREASE_the_overlap_a_hurdle_and_the_concrete_test_2026-07-20]], [[Keeper_K780_de_inflation_ratify_Lyra_catches_correct_my_own_framing_reanchor_on_Q1_2026-07-20]].